incoai/GLM-5.3-DFlash2

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GLM-5.3-DFlash2

Blog | GitHub

This repository contains the DFlash 2 draft model for `zai-org/GLM-5.3`. It is not a standalone language model: it runs inside a speculative decoding server and drafts tokens for the target model to verify.

DFlash 2 is a block-diffusion drafter for speculative decoding. It predicts a whole block of tokens in a single pass and keeps the top candidates at every position. A lightweight selector then traces one coherent path through them. Two-tap dynamic convolutions in the backbone keep the draft from decaying toward the end of the block. Decoding is lossless: greedy output matches the target model exactly, and sampling preserves its distribution.

Downloads last month
3,112

Safetensors

Model size	2B params
Tensor type	BF16

Files info

Inference Providers NEW

Text Generation

This model isn't deployed by any Inference Provider.

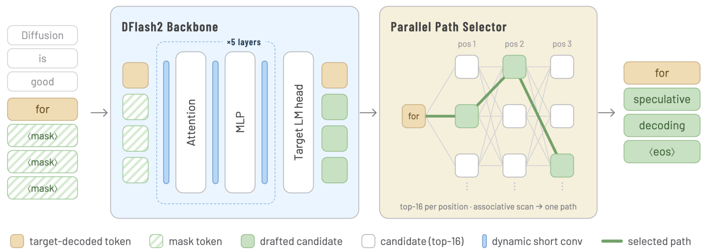
🙋 Ask for provider support

Model tree for `incoai/GLM-5.3-DF…`

Base model `zai-org/GLM-5.3`
Finetuned (3) this model
Quantizations 2 models

Collection including `incoai/GLM-5…`

DFlash 2 Collection



Quick Start

Serve with SGLang:

```
pip install "sglang[all] @ git+https://githul

sglang serve \
  --model-path zai-org/GLM-5.3 \
  --tp-size 4 \
  --trust-remote-code \
  --speculative-algorithm DFLASH \
  --speculative-draft-model-path incoai/GLM-!
  --speculative-draft-attention-backend fa4
```

DFlash 2 is also supported by vLLM v0.28.0 and later; see `incoai/GLM-5.3-NVFP4` for a vLLM serving example with the NVFP4-quantized target. See the blog post for more details.

Evaluation

- Runtime: SGLang on four NVIDIA GB300 GPUs (TP4), with FlashAttention 4 for DFlash 2 draft attention
- Speculation block size: 8 (7 draft tokens per verification step)
- Sampling: GLM-5.3's officially recommended parameters (temperature 1.0, top-p 0.95), with the default `Max` reasoning effort
- Maximum new tokens: 4096

- Samples: 128 at concurrency 1; 1,024 at concurrency 8 and 32

We compare autoregressive decoding, GLM-5.3's native MTP, and DFlash 2. All speculative methods propose seven draft tokens per verification step.

Acceptance Length

Acceptance length is the per-request mean of completion tokens divided by verification steps. Higher is better.

Task	MTP	DFlash 2
GSM8K	5.12	**5.94**
MATH-500	5.05	**6.02**
HumanEval	4.85	**5.48**
MBPP	4.34	**4.95**
MT-Bench	3.81	**4.19**

Throughput

Throughput is total output tokens divided by end-to-end wall time. Each cell shows `output tok/s (speedup vs. autoregressive)`.

Concurrency 1

Task	Autoregressive	MTP	DFlash 2
GSM8K	113.4	292.6 (2.58×)	**366.6 (3.23×)**

Task	Autoregressive	MTP	DFlash 2
MATH-500	113.1	297.4 (2.63×)	**383.3 (3.39×)**
HumanEval	113.7	292.9 (2.58×)	**363.7 (3.20×)**
MBPP	113.4	266.5 (2.35×)	**336.5 (2.97×)**
MT-Bench	113.2	206.8 (1.83×)	**244.3 (2.16×)**

Concurrency 8

Task	Autoregressive	MTP	DFlash 2
GSM8K	535.3	1,094.5 (2.04×)	**1,310.4 (2.45×)**
MATH-500	549.6	1,145.6 (2.08×)	**1,409.7 (2.56×)**
HumanEval	554.4	1,133.8 (2.05×)	**1,360.6 (2.45×)**
MBPP	554.2	1,049.7 (1.89×)	**1,277.4 (2.31×)**
MT-Bench	544.5	807.0 (1.48×)	**895.5 (1.64×)**

Concurrency 32

Task	Autoregressive	MTP	DFlash 2
GSM8K	1,142.7	2,283.6 (2.00×)	**2,694.5 (2.36×)**
MATH-500	1,251.8	2,943.2 (2.35×)	**3,559.8 (2.84×)**
HumanEval	1,303.1	3,016.9 (2.32×)	**3,589.1 (2.75×)**
MBPP	1,292.8	2,790.2 (2.16×)	**3,380.4 (2.61×)**
MT-Bench	1,262.7	2,119.5 (1.68×)	**2,345.0 (1.86×)**

License

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Citation

If you find DFlash 2 useful, please cite:

```
@misc{inco2026dflash2,
  title  = {{DFlash 2: Keep Drafting Paralle
  author = {{Inco AI}},
  year   = {2026},
  month  = {August},
  url    = {https://inco.ai/blog/dflash2/}
}
```

Please also cite the original DFlash paper:

```
@inproceedings{chen2026dflash,
  title     = {{DFlash: Block Diffusion for I
  author    = {Chen, Jian and Liang, Yesheng
  booktitle = {International Conference on Ma
  year      = {2026}
}
```